

Public 311 data alone cannot answer SLA timing questions.

This working prototype uses clearly labeled synthetic records to show the decision workflow.

The buyer problem: public 311 exports can show creation date, status, ward, request type, division, and section, but Toronto's current public file does not expose target or closure timestamps. An honest SLA workflow must join authorized internal work-order fields before it can calculate timing outcomes.

The useful result: an adapter-ready calculation engine that normalizes enriched timestamps, applies configurable response rules, computes margins, and creates a review queue. The bundled proof uses synthetic inputs so its behavior is reproducible without implying official municipal results.

15/15

synthetic labels reproduced

40.0%

scenario ward rate

5

simulated breaches

1

data review isolated

Actual working-output screenshot: synthetic SLA scenario

01 / ADJUST SLA THRESHOLD RULES

Category Target Days

Pothole Repair (Days)

5

Streetlight Maintenance (Days)

3

Tree Trimming (Days)

10

Garbage Collection (Days)

2

Evaluate SLA Engine ->

Reset Standard Rules (5/3/10/2)

Apply Strict Rules (3/2/5/1)

Synthetic SLA scenario inspired by Toronto's licensed public 311 schema. The public export does not include target or closure timestamps. Zero personal data processed.

02 / OPERATIONAL SUMMARY

15 Records Evaluated

Executed python sla_engine.evaluate_dataset in 1.007 ms. Reference now: 2026-08-08T12:00:00+00:00

8

COMPLIANT

1

AT RISK

5

SLA BREACHED

1

DATA REVIEW

5 BREACHED

03 / WARD BOTTLENECK ANALYTICS

Ward Compliance Rankings

WARD NAME	TOTAL REQUESTS	COMPLIANT	AT RISK	BREACHED	COMPLIANCE RATE
Ward 01 - Etobicoke North (Bottleneck)	5	2	0	3	40%
Ward 14 - Toronto-Danforth	5	3	1	1	60%
Ward 10 - Spadina-Fort York	5	3	0	1	75%

04 / EVALUATED SERVICE REQUEST QUEUE

Filtered Records & Audit Reasons

SR-311-001 - Pothole Repair

COMPLIANT

Ward: Ward 10 - Spadina-Fort York Created: 2026-08-01T09:00:00+00:00 Target SLA: 5 days (2026-08-06T09:00:00+00:00)

Closed: 2026-08-04T14:30:00+00:00 Duration: 3.23 days Margin: 1.77 days

Reason Code: CLOSED_BEFORE_SLA_TARGET | Lifecycle: CLOSED_ON_TIME

Source context: City of Toronto 311 public schema and Open Government Licence - Toronto. The displayed IDs, categories, SLA rules, target dates, closure dates, and outcomes are synthetic. Zero personal data is used.

Missing public timing fields require an explicit adapter boundary.

The creative obstacle: prove the workflow without inventing official municipal SLA results.

The technical challenge: the official Toronto 2026 export includes creation date and current status, but not target or closure timestamps. Treating absent fields as evidence would create false breach or compliance claims.

The implemented correction: separate the licensed public schema from a synthetic, adapter-ready SLA contract. The proof exercises ISO8601 parsing, configurable hypothetical targets, explicit **INCOMPLETE_DATA_REVIEW** routing, and machine-readable fixture labels. Production use would require authorized internal target and closure fields.

Actual code excerpt: explicit fixture label and failure route

```
closed_dt = parse_iso_timestamp(raw_closed)

if raw_closed not in (None, "") and closed_dt is None:
    return review("INVALID_CLOSED_TIMESTAMP", sla_target_days)
if raw_status not in VALID_STATUSES:
    return review("UNKNOWN_STATUS", sla_target_days)
if raw_status == "CLOSED" and closed_dt is None:
    return review("MISSING_CLOSED_TIMESTAMP", sla_target_days)
if raw_status != "CLOSED" and closed_dt is not None:
    return review("STATUS_CLOSED_DATE_CONFLICT", sla_target_days)
if closed_dt is not None and closed_dt < created_dt:
    return review("CLOSED_BEFORE_CREATED", sla_target_days)
if closed_dt is not None and closed_dt > reference_now:
    return review("CLOSED_AFTER_REFERENCE_TIME", sla_target_days)
if closed_dt is None and reference_now < created_dt:
    return review("CREATED_AFTER_REFERENCE_TIME", sla_target_days)
```

15/15

synthetic labels reproduced

17/17

unit & integration tests passed

1

malformed record isolated

SHA-256

content fingerprinting

Boundary: scenario calculations demonstrate a review workflow. They are not City of Toronto results, legal SLA certifications, production staffing recommendations, or dispatch commands.

Change the rule thresholds. Watch the queue update.

Reproduce the synthetic benchmark, inspect scenario analytics, and export labeled queues.

15

records evaluated

3

city wards analyzed

8/1/5/1

compliant / risk / breach / error

17/17

automated tests passing

Contrasting Rule Scenario: Reducing the hypothetical Pothole Repair target from 5 days to 3 days re-classifies synthetic request SR-311-001 (completed in 3.23 days) from **COMPLIANT** to **SLA_BREACHED**.

Synthetic Ward Scenario Ranking

Ward 10 - Spadina-Fort York | Total: 5 | Eval: 4 | Compliant: 3 | Breached: 1 | Compliance: 75.0%

Ward 14 - Toronto-Danforth | Total: 5 | Eval: 5 | Compliant: 3 | Breached: 1 | Compliance: 60.0%

Ward 01 - Etobicoke North | Total: 5 | Eval: 5 | Compliant: 2 | Breached: 3 | Compliance: 40.0% (BOTTLENECK)

Try it locally or in the web workbench in about one minute:

1. Open the public web demo link below (no login required).
2. Click 'Evaluate SLA Engine' to run default 5/3/10/2 day SLA rules.
3. Click 'Apply Strict Rules (3/2/5/1)' and observe request status re-classifications.
4. Export evaluated JSON / CSV queues with explicit audit reason codes.

[Working web demo: municipal-311-sla-operations-desk.vercel.app](https://municipal-311-sla-operations-desk.vercel.app)

[GitHub Repository, evidence, and local launch instructions](#)